

Data Communications: Transferring data over a transmission medium between two or more devices, systems, or places is known as data communication.

Components: A communication system is made up of the following components:

1.Message: A message is a piece of information that is to be transmitted from one person to another. It could be a text file, an audio file, a video file, etc.

2.Sender: It is a device that sends data messages. It can be a computer, mobile, telephone, laptop, video camera, or workstation, etc.

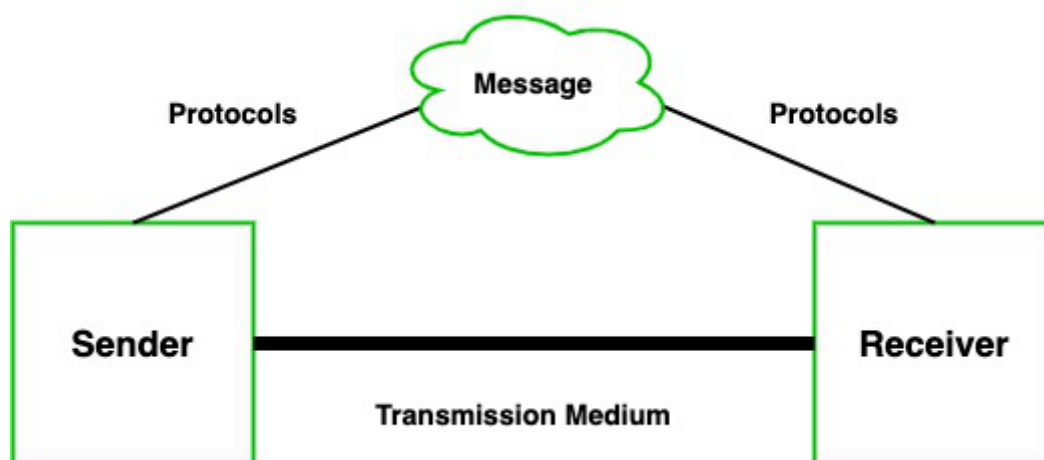
3.Receiver: It is a device that receives messages. It can be a computer, telephone mobile, workstation, etc.

4.Transmission Medium / Communication Channels: Communication channels are the medium that connect two or more workstations. Workstations can be connected by either wired media or wireless media.

5.Set of rules (Protocol): When someone sends the data (The sender), it should be understandable to the receiver also otherwise it is meaningless. Therefore, there are some set of rules (protocols) that is followed by every computer connected to the internet and they are:

i. TCP (Transmission Control Protocol): TCP breaks data into smaller packets before sending and puts them back together in the right order at the destination. It also checks that all packets are delivered correctly.

ii. IP (Internet Protocol): IP takes care of addressing. It makes sure each packet goes to the right device, so your message reaches the correct destination.



Fundamental Characteristics:

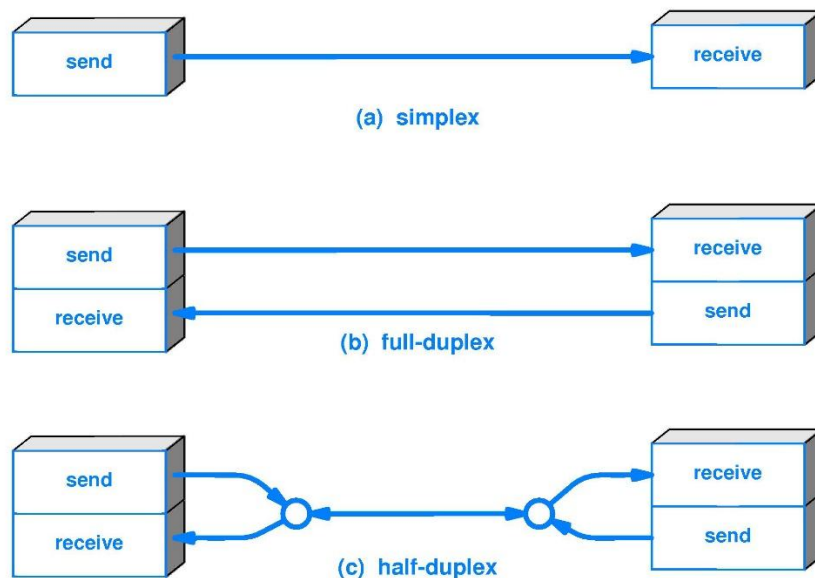
1.Delivery: The data must reach the correct destination device or user. It should be delivered to the right place and to the intended recipient, not anyone else.

2.Accuracy: The data must be delivered without errors. Any changes or corruption during transmission should be detected and corrected.

3.Timeliness: The data should reach the destination on time. In real-time systems like video calls or online gaming, even a small delay can cause problems.

4.Jitter: Jitter refers to the variation in the time delay of received packets. For smooth communication-especially in audio and video streaming-this variation should be minimal.

Type of data communication OR Data Flow: As we know that data communication is communication in which we can send or receive data from one device to another. The data communication is divided into three types:



1.Simplex Communication: It is one-way communication or we can say that unidirectional communication in which one device only receives and another device only sends data and devices use their entire capacity in transmission. For example, IoT, entering data using a keyboard, listening music using a speaker, etc.

2.Half Duplex communication: It is a two-way communication, or we can say that it is a bidirectional communication in which both the devices can send and receive data but not at the same time. When one device is sending data then another device is only receiving and vice-versa. For example, walkie-talkie.

3.Full-duplex (Duplex)communication: It is a two-way communication or we can say that it is a bidirectional communication in which both the devices can send and receive data at the same time. For example, mobile phones, landlines, etc.

Data Representation:

1.Text: Text is stored in computers as a series of bits (like 0001111). Each letter, number, or symbol is given a unique bit pattern. Special codes, such as ASCII or Unicode, are used to change text into binary so computers can understand it.

Text File Formats: .doc,.docx, .pdf, .rtf, .txt, etc.

2.Numbers: Numbers are not stored using ASCII. They are directly changed into binary form (using 0s and 1s) so the computer can do math operations. While people use the base-10 system, computers use base-2 (binary).

Number File Formats: Integer, Fixed point, Date, Boolean, Decimal, etc.

3.Images: Images are stored as bit patterns too. An image is made up of tiny dots called pixels, arranged in a grid. Each pixel has a value that shows its color or brightness. The picture size depends on its resolution (number of pixels). For a simple black-and-white image, 1 can represent black and 0 can represent white.

Image File Formats: Image can be in the format of jpeg, PNG, TIFF, GIF, etc.

There are three main methods of representing color images:

- i. RGB (Red, Green, Blue):** Each color is formed by combining different intensities of red, green, and blue light. This is the most common method used in monitors, TVs, and cameras.
- ii. CMY/CMYK (Cyan, Magenta, Yellow, [Black]):** Used mainly in color printing. Colors are created by subtracting light using inks of cyan, magenta, and yellow. Black (K) is added for depth and detail.
- iii. HSV/HSI (Hue, Saturation, Value/Intensity):** This method describes colors more like humans perceive them-by their shade (hue), richness (saturation), and brightness (value or intensity). It's often used in image processing and editing.

4.Audio: Audio signal is a representation of sound or music. Audio differs from all i.e. from text, number and images. Audio is a series of binary numbers for digital signals. It is continuous but not discrete.

5.Audio File Formats: MP3, M4A audio file type, FLAC, WAV, WMA, AAC, etc.

Video: Video refers to the recording, broadcasting, copying or playback. Video can either be produced or it is continuous and sometimes it is a combination of multiple images produced in motion.

Video File Formats: MP4, MOV, AVI, FLV, etc.

Networks: A computer network is a system of connected devices, like computers and printers, that can communicate with each other and share resources. These devices, called nodes, are linked by cables or wireless signals to transmit data, enabling collaboration, sharing of information like files and emails, and efficient communication

Criteria: The criteria that have to be met by a computer network are:

1. Performance: It is measured in terms of transit time and response time.

- **Transit time is the time for a message to travel from one device to another**
- **Response time is the elapsed time between an inquiry and a response.**

Performance is dependent on the following factors: The number of users, Type of transmission medium, Capability of connected network, Efficiency of software, Bandwidth, Network topology, Network protocols, Distance, Network congestion, Network hardware.

2. Reliability: It is measured in terms of Frequency of failure, Recovery from failures, Robustness during catastrophe, Quality of service (QoS), Reducing single points of failure, Capacity planning, Network architecture.

3. Security: It means protecting data from unauthorized access.

4.Network topology: It is another crucial factor to consider when designing a computer network. It refers to the way in which computers, devices, and links are arranged in a network. Common topologies include bus, star, ring, mesh, and hybrid, each with its own advantages and disadvantages in terms of cost, scalability, reliability, and performance.

Physical Structure: The physical structure of a computer network refers to how the computers and devices are connected. It mainly includes:

1.Type of Connection:

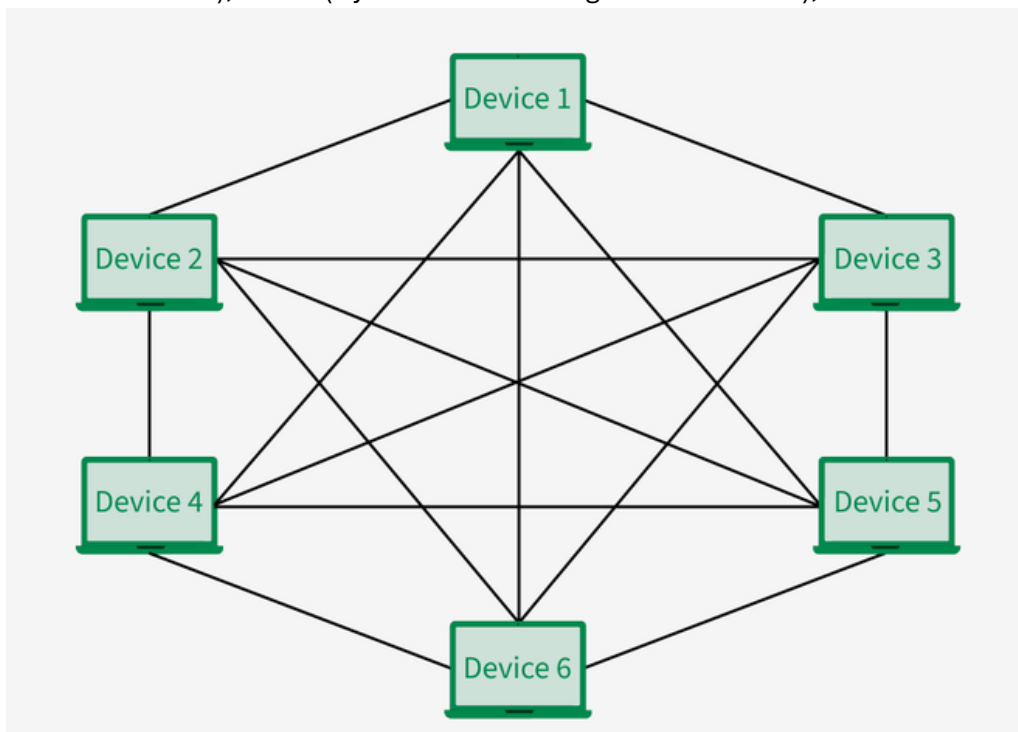
- **Point-to-Point:** A direct link between two devices.
- **Multipoint:** A single link shared by multiple devices.

2. Network Topology: This describes the physical layout or arrangement of devices in the network. Common types are:

- **Bus Topology:** All devices share a single communication line.
- **Star Topology:** All devices are connected to a central hub or switch.
- **Ring Topology:** Each device is connected to two others, forming a ring.
- **Mesh Topology:** Every device is connected to every other device.

Topology: A network topology is the arrangement of devices (nodes) and connections (links) in a computer network. It shows how computers, servers, and other devices are connected and how data flows between them.

1. Mesh Topology: In a mesh topology, every device is connected to another device via a particular channel. Every device is connected to another via dedicated channels. These channels are known as links. In Mesh Topology, the protocols used are AHCP (Ad Hoc Configuration Protocols), DHCP (Dynamic Host Configuration Protocol), etc.



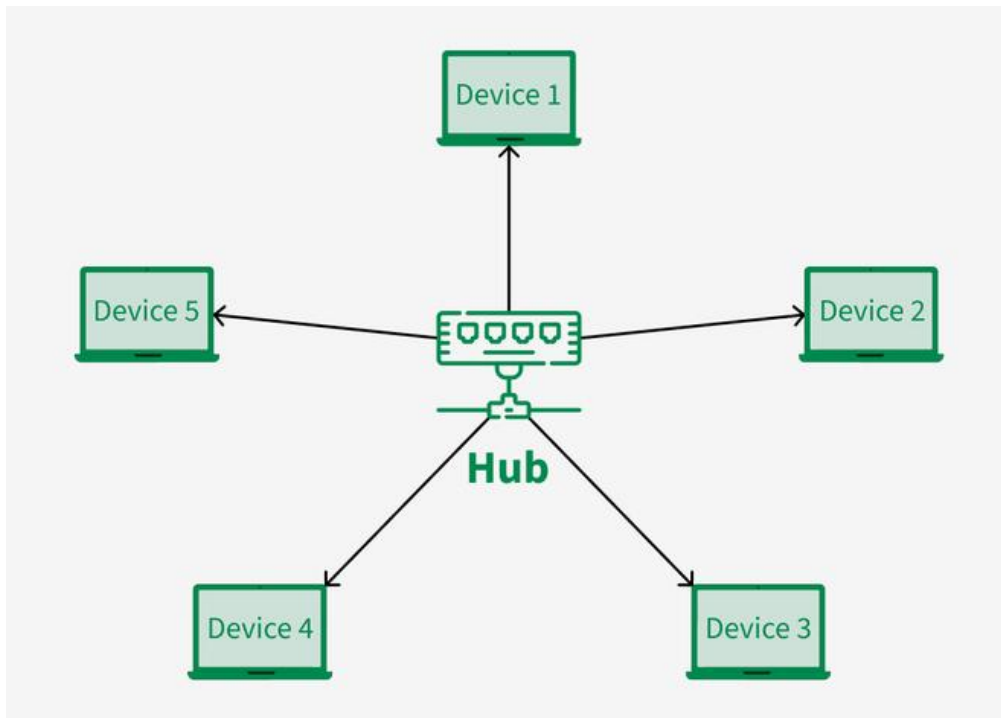
Advantages of Mesh Topology:

- Communication is very fast between the nodes.
- Mesh Topology is robust.
- Provides security and privacy.

Disadvantages of Mesh Topology:

- Installation and configuration are difficult.
- The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.
- The cost of maintenance is high.

2.Star Topology: In Star Topology, all the devices are connected to a single hub through a cable. This hub is the central node and all other nodes are connected to the central node. The hub can be passive in nature i.e., not an intelligent hub such as broadcasting devices, at the same time the hub can be intelligent known as an active hub. Active hubs have repeaters in them.



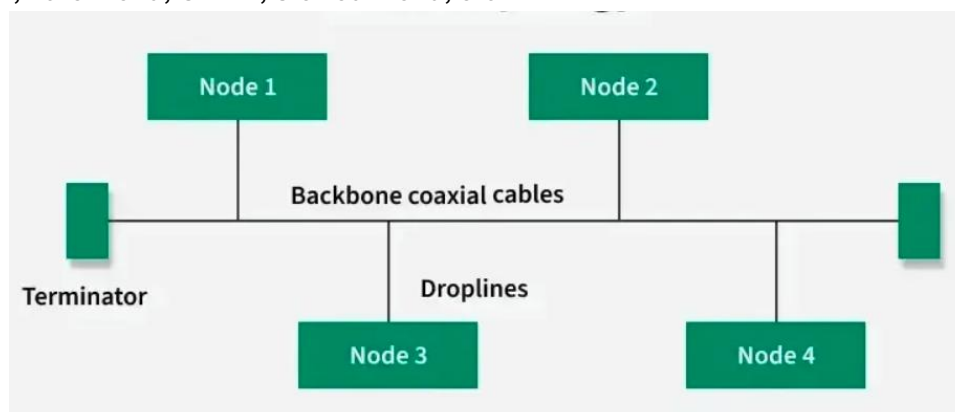
Advantages of Star Topology:

- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.
- Star topology is cost-effective as it uses inexpensive coaxial cable.

Disadvantages of Star Topology:

- If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.
- The cost of installation is high.

3.Bus Topology: Bus Topology is a network type in which every computer and network device is connected to a single cable. It is bi-directional. It is a multi-point connection and a non-robust topology because if the backbone fails the topology crashes. In Bus Topology, various MAC (Media Access Control) protocols are followed by LAN ethernet connections like TDMA, Pure Aloha, CDMA, Slotted Aloha, etc.



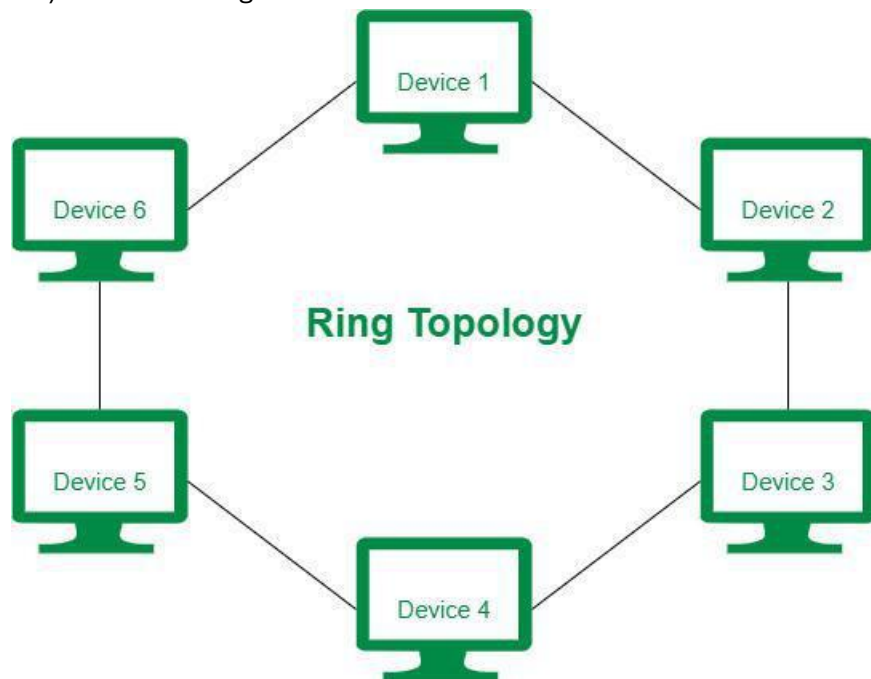
Advantages of Bus Topology:

- It's simple to install and doesn't require much cabling.
- Uses less cable compared to other topologies, so it's cheaper.
- New devices can be added easily by joining them to the main cable.

Disadvantages of Bus Topology:

- A bus topology is quite simpler, but still, it requires a lot of cabling.
- If the common cable fails, then the whole system will crash down.

4. Ring Topology: In a ring topology, each device (computer or node) is connected to exactly two other devices, forming a circular path for data to travel. Data moves in one direction (or sometimes both) around the ring until it reaches its destination.

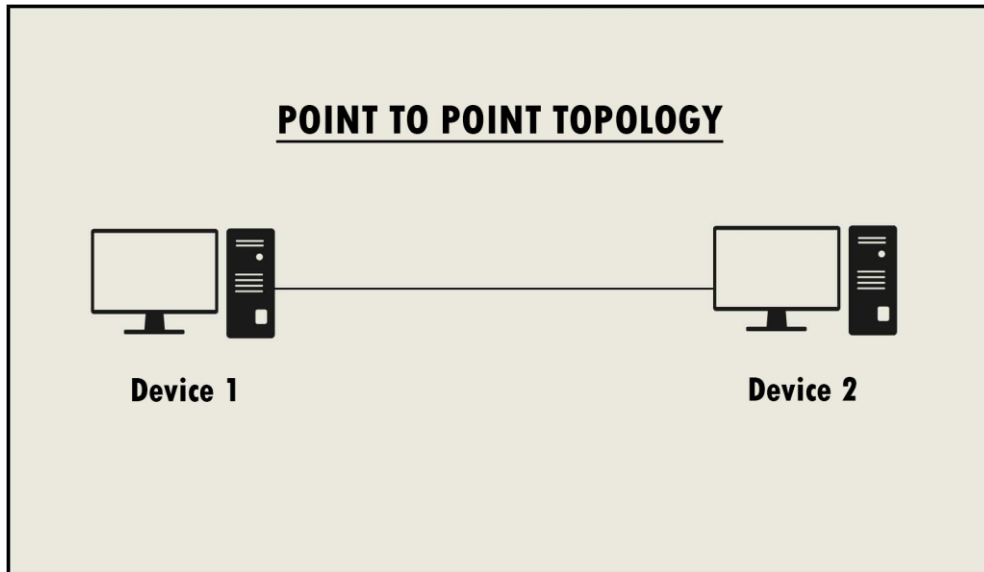
**Advantages of Ring Topology:**

- The data transmission is high-speed.
- The possibility of collision is minimum in this type of topology.
- Cheap to install and expand.

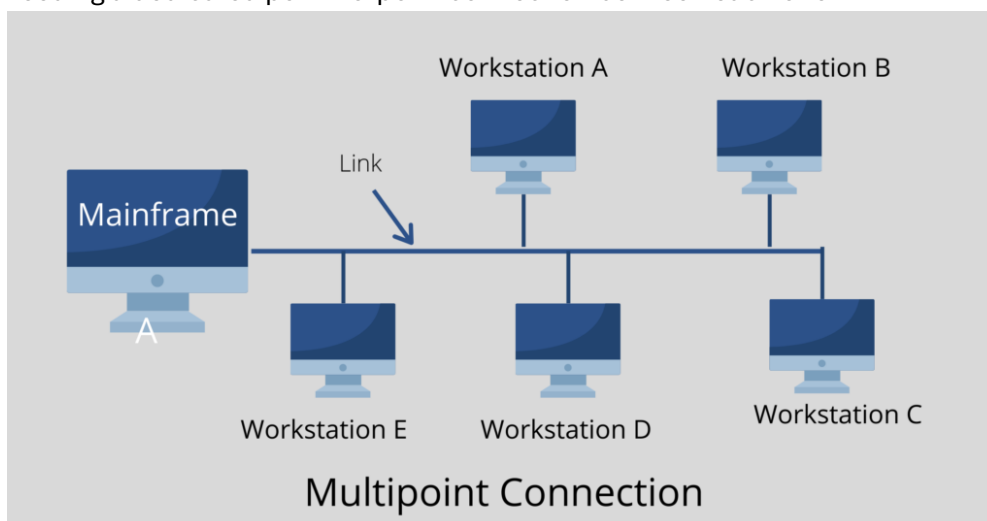
Disadvantages of Ring Topology:

- The failure of a single node in the network can cause the entire network to fail.
- Troubleshooting is difficult in this topology.
- Less secure.

Point to point: Point-to-point topology is the simplest network configuration, connecting just two devices directly with a single, dedicated communication link. This creates a private, reliable, and high-speed connection where the entire bandwidth is available to the two devices.

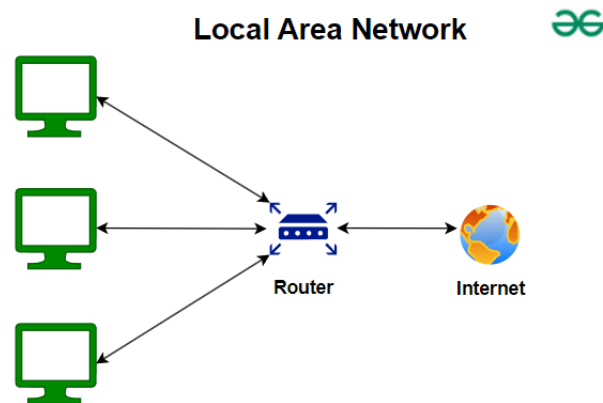


Multi-point: A multipoint topology is a network architecture where multiple devices share a single communication channel or link. It is a cost-effective way to connect many devices without needing a dedicated point-to-point connection between each one.

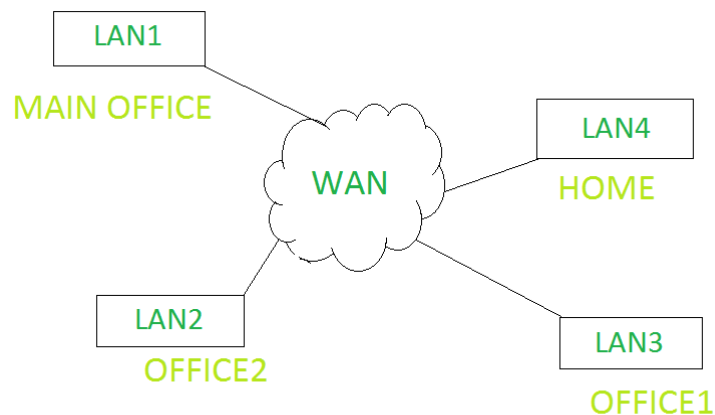


LAN:

- 1.A Local Area Network (LAN) is a network that interconnects computers and devices within a limited geographical area such as a building, campus or office.
- 2.It provides high data transmission speeds because the distance covered is short and the communication medium is usually reliable.
- 3.LANs support resource sharing, allowing users to share hardware, software, data and network services efficiently.
- 4.They are easy to design, install and maintain, and they use network topologies like bus, star or ring to structure the communication among devices.

**WAN:**

1. A Wide Area Network (WAN) is a communication network that covers a large geographical area, often spanning cities, states or even countries.
2. It connects multiple smaller networks, such as LANs and MANs, using long-distance communication links like leased lines, satellite links or fiber optics.
3. WANs are usually created and run by telecom companies or service providers because they require expensive infrastructure.
4. The backbone of a WAN consists of high-speed links like fiber optics, microwave links or satellites.
5. WANs usually have lower data transmission speeds compared to LANs because of the long distances and diverse communication technologies involved.
6. They are used by large organizations, banks and government agencies to enable communication and data exchange across distant locations.

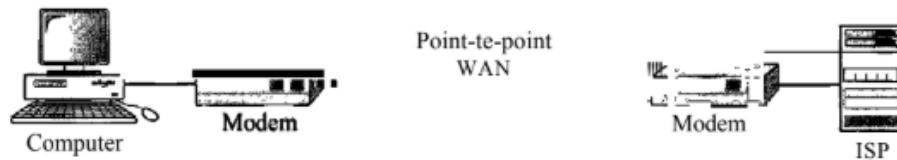
**Point-to-point WAN:**

Description: A single, dedicated, leased line connection between two specific points (e.g., two office locations).

Best for: Connecting just two sites, offering enhanced security and reliability because the connection isn't shared.

Example: A business leasing a private line to connect its headquarters to a single branch office.

Protocols: Uses protocols like PPP (Point-to-Point Protocol) to manage the connection.

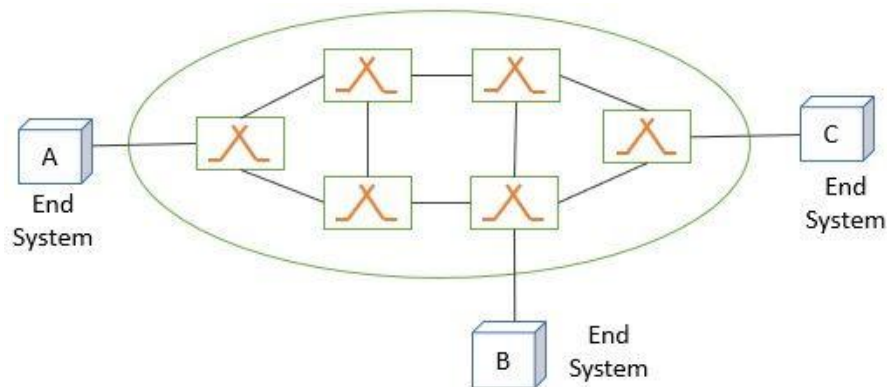


Switched WAN:

Description: A more complex network that connects multiple end nodes (sites) through a mesh of point-to-point connections, sharing a common infrastructure.

Best for: Connecting numerous locations, providing greater efficiency, flexibility, and scalability.

Example: A network that uses a central exchange to route traffic between many different offices or to the internet, like a cellular network or a corporate network connecting multiple branches.



Internet: The Internet is a global network of interconnected computers and devices that allows users to access and share information and devices, allowing them to communicate and exchange data. It enables users to access a wide range of services, such as websites, emails, social media, online applications, and cloud storage. Through standardized communication protocols, the Internet facilitates the sharing of information, resources, and media across vast distances, making it an essential tool for worldwide:

- Education
- Business
- Entertainment
- Personal Communication

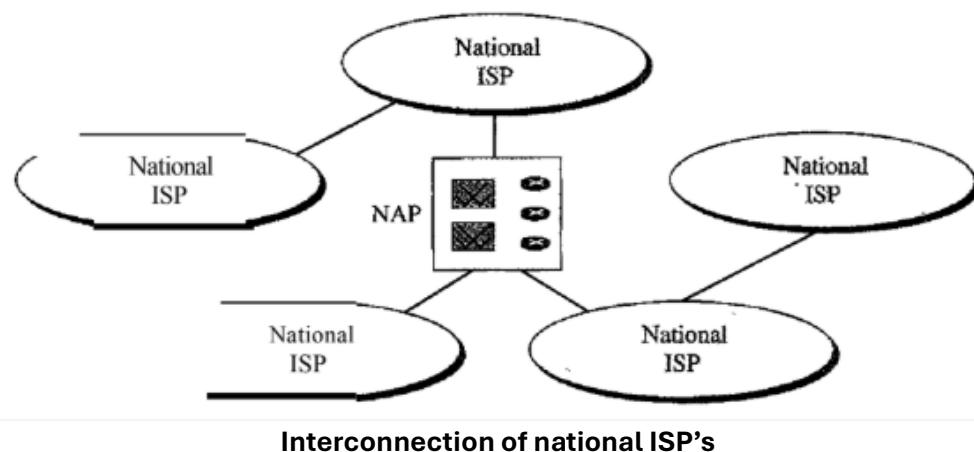
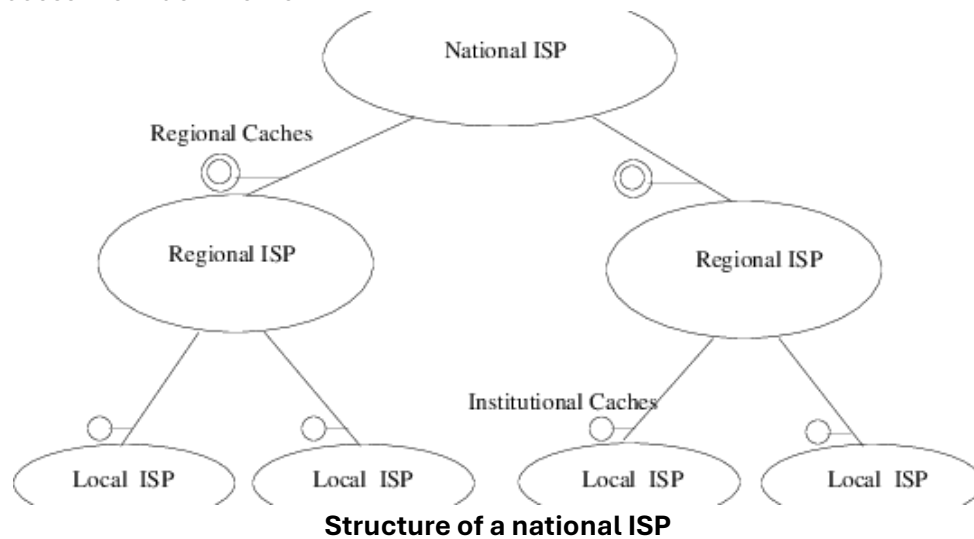
ISP Hierarchy:

1. International Internet Service Providers: These are the top-level ISPs. They connect different countries and continents. They form the global backbone of the internet. All other ISPs depend on them for worldwide connectivity.

2. National Internet Service Providers: National ISPs maintain large backbone networks within a country. They connect to international ISPs through network access points (NAPs). They may also connect with each other through private switching stations called peering points. They usually operate at very high data rates.

3. Regional Internet Service Providers: Regional ISPs are smaller than national ISPs. They connect to one or more national ISPs. They operate at lower data rates and serve specific regions or states.

4. Local Internet Service Providers: Local ISPs provide internet service directly to end users such as homes, offices, colleges or small organizations. They connect to regional or national ISPs to access the wider internet.



The Internet (1960's): The Internet is a global network that connects millions of computers and devices, enabling people to communicate, share information, and access digital resources worldwide. It works like a vast highway system, where data travels in small packets between connected devices.

- Facilitates instant communication and information exchange across the world.
- Connects computers, smartphones, and servers through various wired and wireless networks.
- Powers daily activities such as emailing, online learning, shopping, and entertainment.

Standards and Administration of the Internet:

1. Internet Standards: Internet standards are rules and guidelines that make sure all devices and networks on the internet can work together. These standards define how data is formatted, transmitted and received. Protocols like TCP/IP, HTTP, SMTP and DNS are examples of internet standards.

2. Who Creates These Standards: The standards are developed by recognized organizations, mainly:

- IETF (Internet Engineering Task Force): Creates technical standards such as TCP/IP.
- W3C (World Wide Web Consortium): Develops standards for the web, like HTML and CSS.
- ISO (International Organization for Standardization): Creates general international standards.

3. Administration of the Internet: No single organization controls the entire internet. Instead, different groups share responsibilities:

- ICANN (Internet Corporation for Assigned Names and Numbers): Manages domain names and IP address allocation.
- IANA (Internet Assigned Numbers Authority): Handles global coordination of IP addresses and protocol numbers.
- Regional Internet Registries (RIRs): Assign IP addresses to local ISPs in different regions.

4. Purpose: These organizations make sure the internet runs smoothly, follows rules and stays secure, reliable and globally accessible.

Protocol: A protocol is a set of rules that controls how data is sent and received over a network. It decides how devices communicate, how messages are formatted, how errors are handled and how data reaches the correct destination. **Ex:** Include TCP/IP, HTTP, FTP and SMTP.

Protocol Layering: Protocol layering is a way of organizing network communication into separate layers, where each layer has a specific function. Instead of one large, complex system, the network is divided into smaller layers. Each layer handles its own task and works with the layers above and below it. The change completed in one layer, then it affects the next layer. **Ex:** One layer might handle physical transmission, another handles routing and another manages applications.

Basic elements of layer architecture:

1. Services: Each layer provides certain functions or services to the layer above it. These services help the upper layers perform their tasks.

2. Protocols: A protocol defines the rules each layer follows for communication. Devices on the same layer use the same protocol to exchange data.

3. Interfaces: Interfaces define how one layer interacts with the layer below it. They specify what information is passed between layers.

OSI Model:

1.Physical Layer: First layer is the physical layer, which deals with the physical aspects (such as cables, connectors, and signaling) of transmitting data between two devices. This layer define transmission mode also whether it is simplex, half duplex or full duplex. It define topology configuration.

2.Data Link Layer: The second layer is the data link layer, which is responsible for establishing and maintaining communication between two devices on the same network. Data link layer is responsible for moving frames from one hope to next hope.

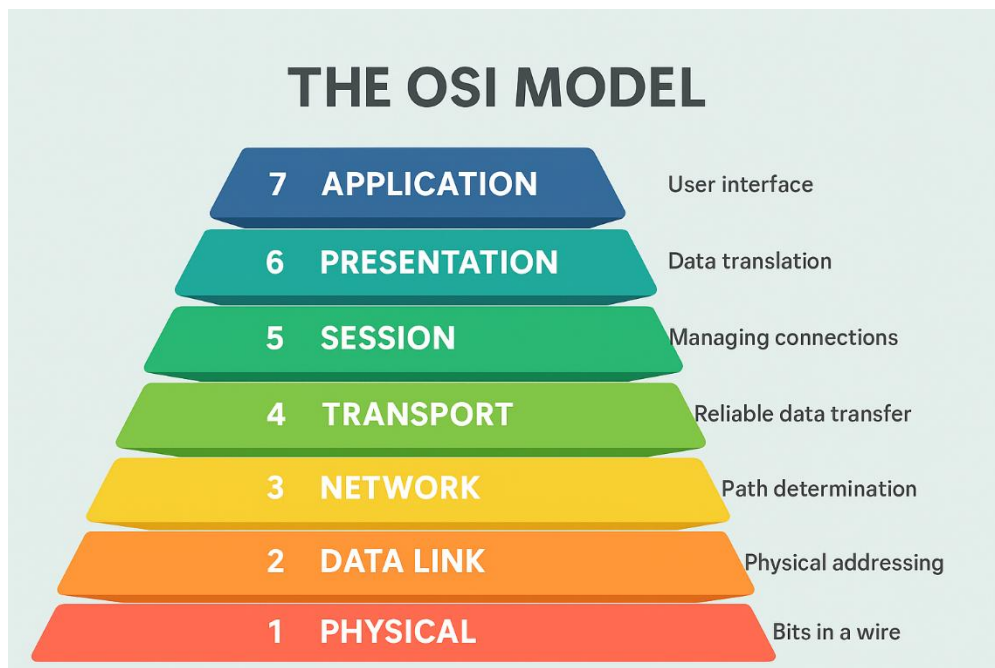
3.Network Layer: The third layer is the network layer, which handles the routing of data between networks. Network layer is responsible for the delivery of individual packet from source to destination.

4.Transport Layer: The fourth layer is the transport layer, which ensures reliable transmission of data between two devices. Transport layer is responsible for process-to-process delivery. A process is an application program running on a host.

5.Session Layer: The fifth layer is the session layer it is also known as network dialog controller, which establishes and manages sessions between applications.

6.Presentation Layer: The sixth layer is the presentation layer, which deals with the syntax and semantics of the data being transmitted.

7.Application Layer: Seventh layer is the application layer, which provides services to applications for network communication.



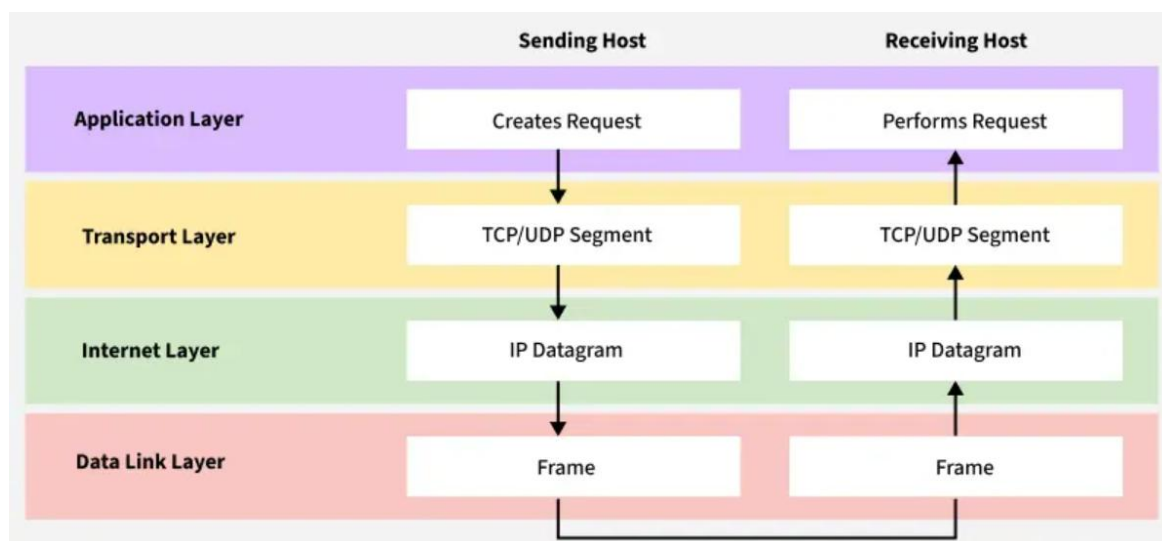
TCP/IP Model:

1.Application Layer: Provides network services to user applications. Protocols like HTTP, FTP, SMTP and DNS work here. It handles functions such as file transfer, email and web access.

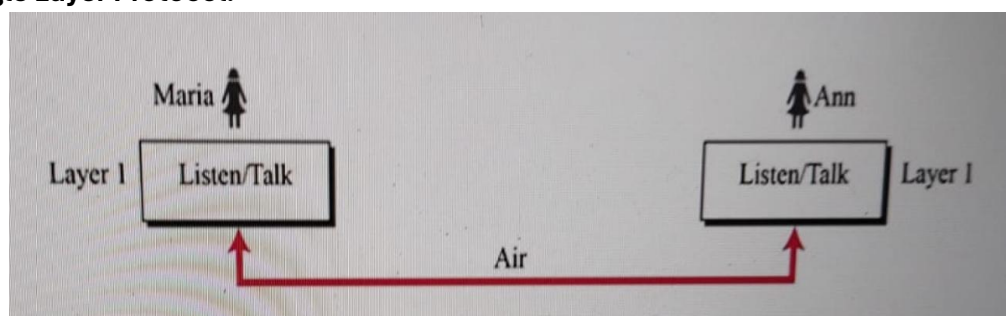
2.Transport Layer: Manages end-to-end communication between devices. It uses TCP for reliable delivery and UDP for faster, connectionless communication. It handles error control and flow control.

3.Internet Layer: Deals with logical addressing and routing of packets across networks. The IP protocol works here. It selects the best path for data to travel.

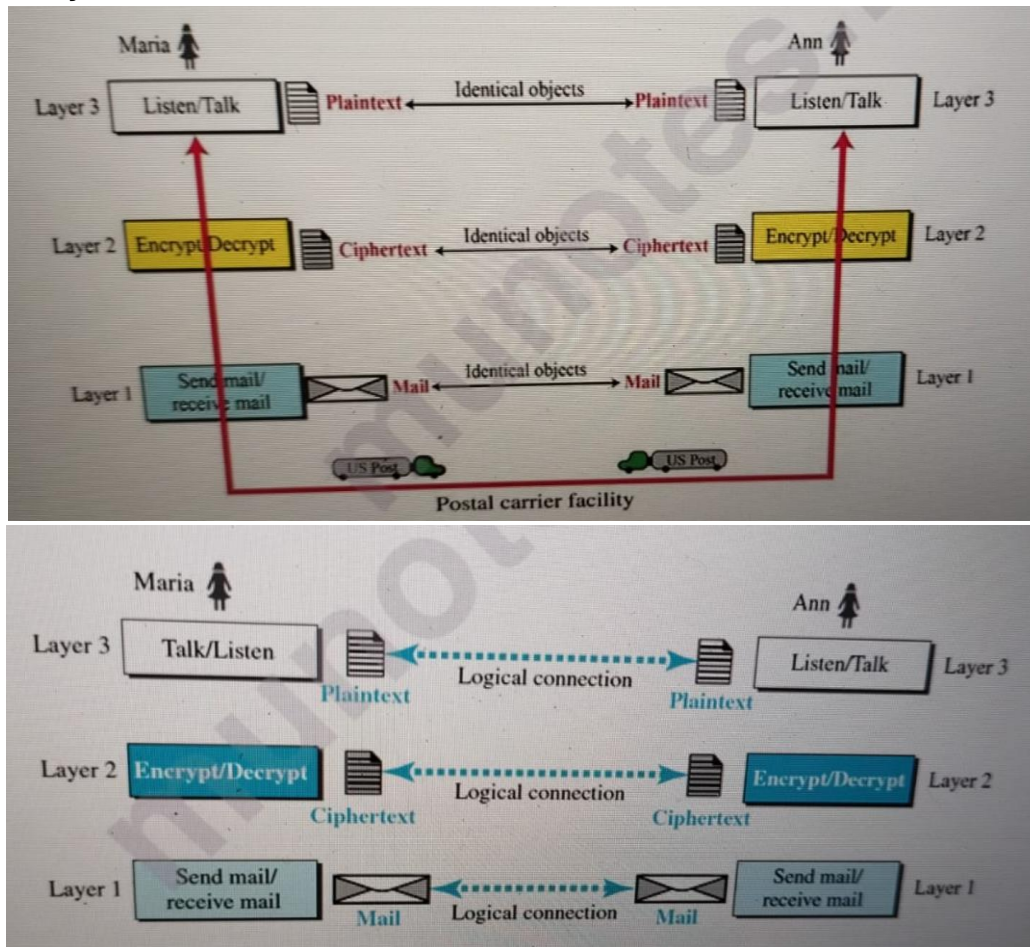
4.Network Access Layer: Handles the physical transmission of data over the network. It includes technologies like Ethernet and Wi-Fi and defines how devices access the communication medium.

**Reasons of layering:**

1. Layering of protocols provides well-defined interfaces between the layers, so that a change in one layer does not affect an adjacent layer.
2. The protocols of a network are extremely complicated and designing them in layers makes their implementation more feasible.

Scenario:**A. Single Layer Protocol:**

1. Maria speaks using the Listen/Talk layer.
2. Her voice travels through the medium, which is air.
3. There are no separate layers to check errors, control flow or manage addresses.
4. Ann listens using her Listen/Talk layer.
5. If Ann hears the message clearly, communication is successful. If noise interferes, there is no mechanism to correct it.

B. Three Layer Protocol:

1.Layer 3: Listen/Talk (Application Layer): Maria creates the message in plain text using the Listen/Talk layer. At Ann's side, the same layer is used to read and understand the message.

2.Layer 2: Encrypt/Decrypt (Security Layer): The plain text message is passed to the Encrypt/Decrypt layer. Here, the message is converted into cipher text to keep it secure. At the receiver's side, Ann's Encrypt/Decrypt layer converts the cipher text back into plain text.

3.Layer 1: Send Mail/Receive Mail (Transport Layer): The encrypted message is then passed to the Send Mail layer, which packages the message and sends it through the postal carrier facility. At Ann's side, the Receive Mail layer collects the message and passes it upward.

4.Flow of Communication: The message moves downward through the layers at the sender, travels through the communication medium (postal system), and then moves upward through the layers at the receiver. Each layer communicates with its corresponding layer using identical objects.

Principle of protocol layering:

1. For two-way (bi-directional) communication, each layer must perform opposite functions at the sender and receiver. One side sends or applies an operation, and the other side receives or reverses it.

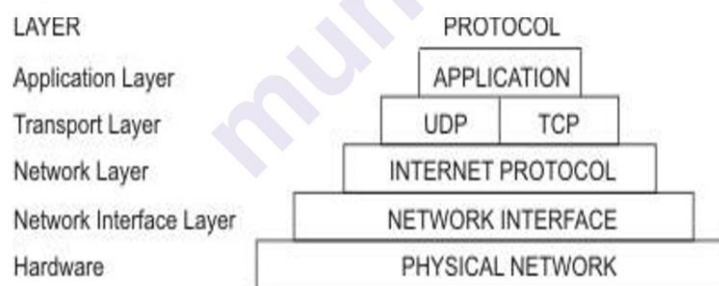
2. Each layer has a fixed role:

- The third layer is responsible for talking and listening.
- The second layer handles encryption and decryption of the message.
- The first layer takes care of sending and receiving the message over the network.

3. The type of data remains the same at corresponding layers on both sides. The third layer uses plain text, the second layer works with encrypted (cipher) text, and the first layer handles data ready for transmission.

4. There is a logical (virtual) connection between the same layers at the sender and receiver. This is called layer-to-layer communication.

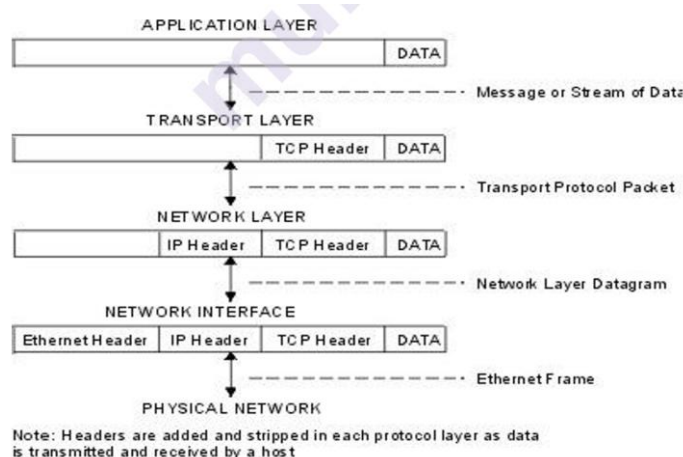
5. Because of this logical connection, the sender and receiver behave as if each layer communicates directly with its matching layer, which makes communication simple and well-organized.

TCP/IP Protocol Suite:

1. Protocols and Layering: Protocols are rules that define message format and communication methods. The TCP/IP protocol suite is organized into layers so that data communication becomes systematic and understandable.

2. Data Flow through Layers: Application programs send data to the Transport layer (TCP or UDP), which divides data into packets and adds addressing information. These packets are passed to the Internet layer.

3. Transmission over Network: The Internet layer wraps packets into IP datagrams and decides the route. The Network Interface layer then sends these datagrams as frames over physical networks like Ethernet.



Layer	Sender Side (Encapsulation)	Receiver Side (Decapsulation)
Application Layer	Application creates the message or data to be sent.	Application receives the final data and understands it.
Transport Layer (TCP)	TCP divides data into segments and adds a TCP header (port numbers, sequence info).	TCP removes the TCP header, checks order and reliability, and passes data upward.
Network Layer (IP)	IP adds an IP header with source and destination IP addresses, forming a datagram.	IP removes the IP header and forwards the packet to the transport layer.
Network Interface Layer	Ethernet adds a frame header (MAC addresses) and sends data over the physical medium.	Ethernet removes the frame header and passes the datagram to the IP layer.
Physical Network	Bits are transmitted over the wire or wireless medium.	Bits are received from the medium and converted back to frames.

IPv4 (Internet Protocol Version 4):

1. IPv4 stands for Internet Protocol Version 4. It is the most widely used protocol for identifying devices on a network. IPv4 provides a unique address to each device so that data can be correctly sent and received over the internet.

2. An IPv4 address is 32 bits long, divided into four parts called octets. Each octet contains 8 bits and can represent values from 0 to 255. The four octets are written as decimal numbers separated by dots, known as dotted decimal notation, for example: 192.168.1.1.

3. Each IPv4 address is divided into different parts:

- **Network Part:** Identifies the network to which the device belongs. All devices in the same network share the same network part.
- **Host Part:** Identifies a specific device within the network. This part must be unique for every host in the network.
- **Subnet Number:** An optional part used to divide large networks into smaller subnets for better management and performance.

4.Types of IPv4 Addressing: IPv4 supports three addressing modes:

- **Unicast:** One sender communicates with one receiver (example: accessing a website).
- **Broadcast:** One sender sends data to all devices in the same network.
- **Multicast:** One sender sends data to a selected group of devices (example: audio or video streaming).

IPv6 (Internet Protocol Version 6):

1. IPv6 is the upgraded version of Internet Protocol, introduced to overcome the address limitation of IPv4 and to support the rapid growth of internet-connected devices.

2. IPv6 uses a 128-bit address space, providing a very large number of unique IP addresses. IPv6 addresses are written in hexadecimal format, divided into 8 groups, where each group represents 16 bits (2 bytes). The groups are separated by colons (:), and each hexadecimal digit represents 4 bits.

3. IPv6 supports the following addressing methods:

- **Unicast:** Communication between one sender and one receiver.
- **Multicast:** Data sent to a group of devices at the same time.
- **Anycast:** Data delivered to the nearest device among multiple devices sharing the same address.

4.IPv6 was introduced mainly to solve IPv4 address exhaustion and to improve network performance. It provides a better header structure for faster routing, supports real-time multimedia traffic using traffic class and flow label fields, and offers enhanced security through built-in authentication and encryption.

5.Thus, IPv6 ensures scalability, efficiency and security for modern internet communication.

MAC Address:

1.A MAC address is a unique physical address assigned to the Network Interface Card (NIC) of a device. It is used to identify devices within a local network and helps in proper delivery of data frames.

2.A MAC address is 48 bits long and is represented in hexadecimal format, usually written as six groups of two hex digits separated by colons or hyphens, for example 00:1A:2B:3C:4D:5E.

3.The MAC address is divided into two parts. The first part identifies the manufacturer of the NIC (Organizationally Unique Identifier), and the second part uniquely identifies the device.

MAC addresses operate at the Data Link Layer of the network model. They are mainly used in

4.LAN communication to ensure data reaches the correct device.

5.Since MAC addresses are hardware-based, they are generally permanent and provide reliable device identification within a network.

Subnet Mask: A subnet (subnetwork) is a smaller, logical section of a larger network.

It is created by dividing an IP network into multiple parts to improve organization, security and efficient use of IP addresses.

Key points:

- A subnet helps control network traffic by keeping local traffic within the subnet.
- It improves performance by reducing congestion.
- Subnets are created using a subnet mask, which separates the network part and host part of an IP address.
- Large networks (like offices or campuses) use subnets to group devices based on departments or locations.

NIC Cards: A Network Interface Card (NIC) is a hardware component that allows a computer or device to connect to a network. It handles sending and receiving data through wired or wireless communication. Each NIC has a unique MAC address used for device identification.

Types of NIC:

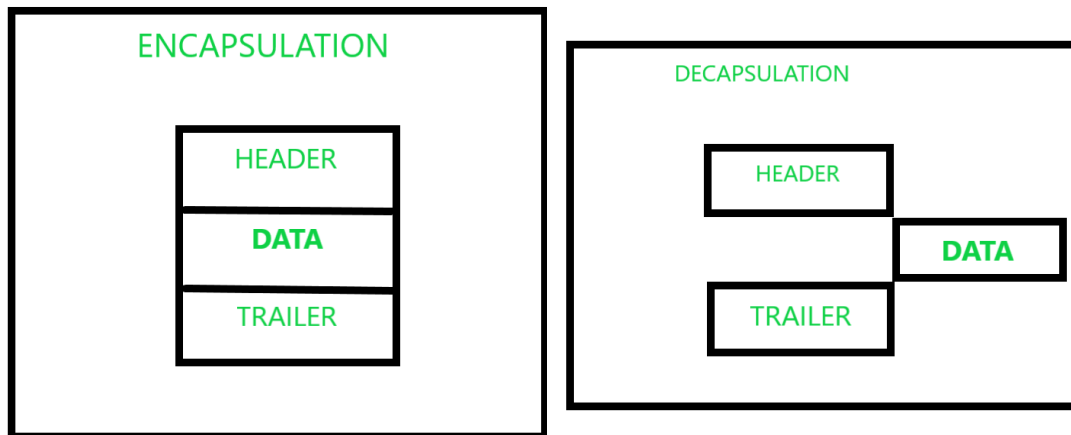
- **Wired NIC:** Uses Ethernet cables (like RJ-45) to connect to the network. Provides stable and high-speed communication.
- **Wireless NIC:** Uses Wi-Fi signals to connect without cables. Allows mobility and easy installation.
- **USB NIC:** A plug-and-play network adapter connected through a USB port. Useful for devices without built-in NICs.
- **Fiber Optic NIC:** Uses fiber optic cables for very high-speed communication over long distances.

Characteristics of NIC:

- **Unique MAC Address:** Every NIC has a permanent hardware address for identification.
- **Data Transmission:** Sends and receives data frames over the network.
- **Compatibility:** Designed for specific network types like Ethernet, Wi-Fi or fiber.
- **Speed Support:** Supports different data rates such as 10 Mbps, 100 Mbps, 1 Gbps or higher.

- **Built-in or External:** Can be integrated into the motherboard or added as an external card.

Encapsulation and Decapsulation:



Encapsulation: Encapsulation refers to attaching new information in the Application Layer data as it is passed onto the next layers in the TCP/IP model. This additional information basically divided into two parts, Header and Trailer. These are elements attached in order to make the transmission more smoother, on each layer a PDU (Protocol Data Unit) is generated.

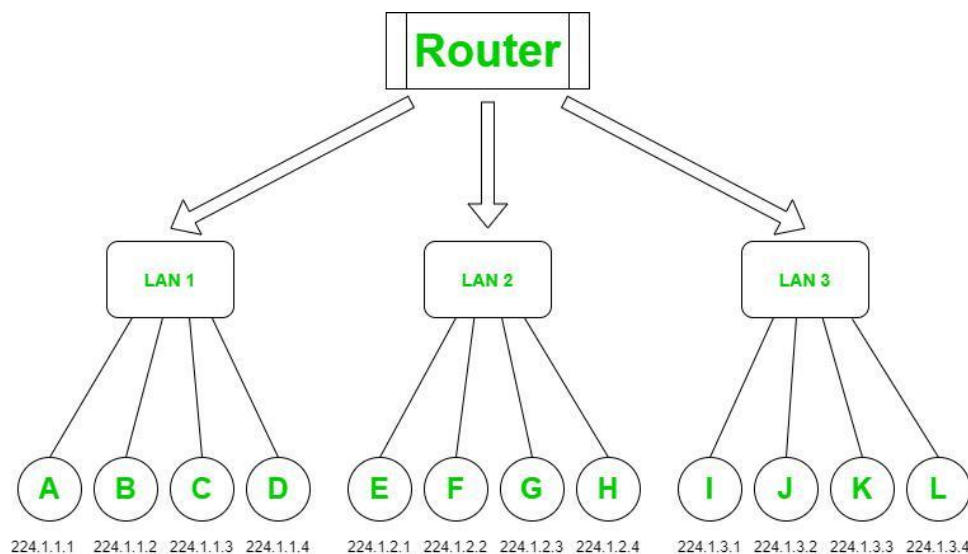
Decapsulation: Decapsulation refers to the removal of all these additional information and extraction of originally existing data, and this process continues till the last layer i.e. the Application Layer. This process removes, fragments of distinct information in each layer as it approaches that layer. Here is the pictorial representation of the whole process.

Encapsulation	Decapsulation
The data moment starts from the upper layer and terminates finally on the lowest layer.	Whereas, here the data moves from the lower layer till the upper layer.
The process involves addition of header and trailer section.	This process involves removal of header and trailer sections
This process executes first and is followed by decapsulation.	This process executes once encapsulation is finally completed.
It occurs inside the source device.	It occurs inside the destination device.

Addressing: Addressing is the method of identifying the sender and receiver in a network. Every device that sends or receives data must have a unique address so that information reaches the correct destination.

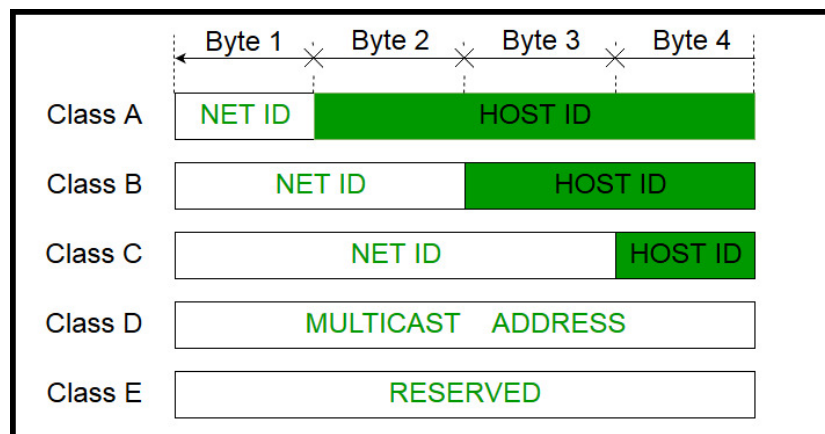
There are three main types of addresses used in communication:

- **Physical Address (MAC Address):** Used at the data link layer. It identifies a device's network interface card. It is fixed and unique.
- **Logical Address (IP Address):** Used at the network layer. It helps identify a device's location in a network and enables routing across multiple networks.
- **Port Address:** Used at the transport layer. It identifies specific applications or processes on a device (like web, email or FTP).



Classful Addressing (Classes of IP Addressing): The 32-bit IP address is divided into five sub-classes. These are given below: Class A, Class B, Class C, Class D and Class E.

1. IP addresses are divided into classes, each with a fixed range, and the first octet determines the class and how bits are split into network ID and host ID.
2. Class D is used for multicast and Class E for experimental purposes.
3. IP addresses are assigned to devices by ISPs or network administrators.



Class A:

- Class A IP addresses are used for networks with a **large number of hosts**.
- In Class A, the **network ID is 8 bits** and the **host ID is 24 bits**.
- The **first bit is always 0**, and the default subnet mask is **255.0.0.0**.
- The number of possible hosts is calculated as $2^{24} - 2 = 16,777,214$.

**Class A****Class B:**

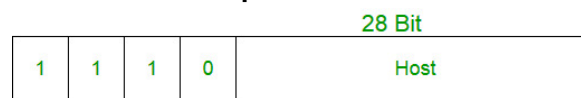
- Class B IP addresses are used for **medium-sized networks**.
- In Class B, the **network ID is 16 bits** and the **host ID is 16 bits**.
- The **first two bits are always 10**, and the default subnet mask is **255.255.0.0**.
- The number of possible hosts is calculated as $2^{16} - 2 = 65,534$.

**Class B****Class C:**

- Class C IP addresses are used for **small networks**.
- In Class C, the **network ID is 24 bits** and the **host ID is 8 bits**.
- The **first three bits are always 110**, and the default subnet mask is **255.255.255.0**.
- The number of possible hosts is calculated as $2^8 - 2 = 254$.

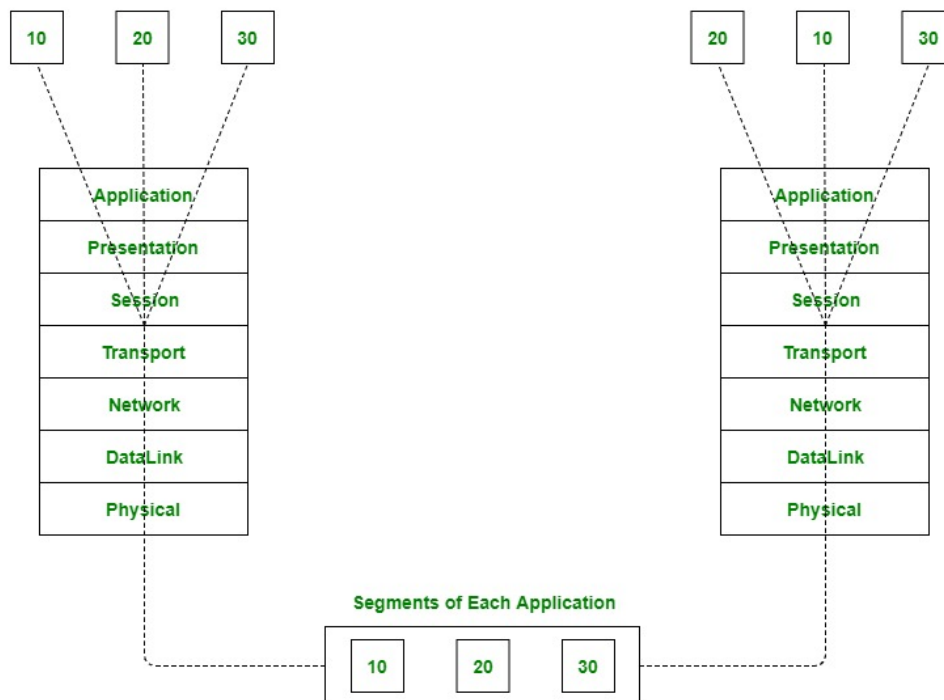
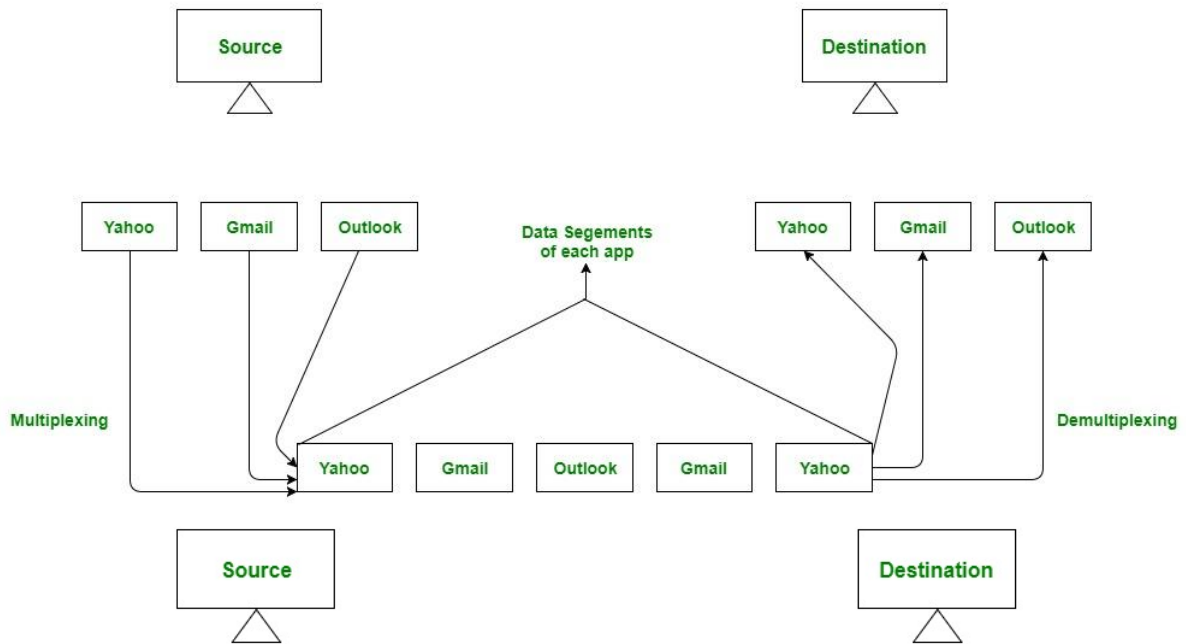
**Class C****Class D:**

- Class D IP addresses are used for **multicast communication**.
- These addresses **do not have network ID and host ID division**.
- The **first four bits are always 1110**.
- They are used to send data to **multiple devices at the same time**.

**Class D****Class E:**

- Class E IP addresses are reserved for **experimental and research purposes**.
- These addresses are **not used for normal communication**.
- The **first four bits are always 1111**.
- They are kept for **future use**.

**Class E**

Multiplexing and Demultiplexing:**Multiplexing / Demultiplexing**

Multiplexing: Gathering data from multiple application processes of the sender, enveloping that data with a header, and sending them as a whole to the intended receiver is called multiplexing.

Demultiplexing: Delivering received segments at the receiver side to the correct app layer processes is called demultiplexing.

How Multiplexing Works:

- Multiple data streams (from different applications or devices) arrive at the sender.
- The multiplexing layer assigns each data stream a unique identifier, such as a port number or channel number.
- It combines all these data streams into one single signal.
- This combined signal is sent over a shared communication channel.

How Demultiplexing Works:

- The combined signal reaches the receiver.
- The demultiplexing layer reads the identifiers (port numbers or channel numbers) attached to each piece of data.
- It separates the data back into individual streams.
- Each stream is delivered to its correct application or device.